MIL-F-0013785B(MU)
24 November 1969
USED IN LIEU OF
MIL-F-13785A
30 November 1955

MILITARY SPECIFICATION

FILTER MATERIAL

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-F-13785A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers several types of particulate filter material.

1.2 Classification. The filter material shall be furnished in the following types, as specified (see 6.2):

Type 5	-	Asbestos-bearing (low resistance)
Type 6	-	Asbestos-bearing (high resistance)
Type 7	-	Asbestos-bearing (high resistance)
Type 8	-	Glass-fiber-bearing (high resistance)
Type 15-N	-	Asbestos and/or glass-bearing (low resistance)

Types 6 and 7 differ in composition (see table 1)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 4240

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SPECIFICATIONS

FEDERAL

O-P-94	-	Paper, Test; pH Indicator.
TT-W-572	-	Wood Preservative, Water-Repellent.
UU-P-31	-	Paper; General Specification and Methods of Testing.
CCC-T-191	-	Textile Test Methods.
MMM-A-260	-	Adhesive, Water-Resistant (For Sealing Waterproofed Paper).
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible.

MILITARY

MIL-B-2427	-	Boxes, Ammunition Packing; Wood, Nailed.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DLB125-8-1	-	Water Repellency Indicator, Q101.
DLB136-42-102-5	-	DOP Filter Testing Penetrometer, Q127.
D150-2-1	-	Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of this publication should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Material. The furnish for the filter sheet shall be made with suitable proportions of the fibers shown in table I.

Table I. Composition

Type	*Fibers
5	b, r, f, h
6	b, e, f, h
7	b, r, a (or i) h
8	g, r, a (or i) h
15-N	b (or g) f, h, r

b - asbestos
r - viscose rayon
f - cotton flock
h - manila hemp
e - esparto
a - hardwood sulfate pulp
i - hardwood sulfite pulp
g - glass fibers (0.4 to 0.6 micron diameter)

*Fibers other than those listed may be used only after test and approval by Edgewood Arsenal.

3.1.1 Gauze backing. Gauze backing is required for type 5 material (see 6.3). The gauze backing shall be treated with at least 2 percent of mildew inhibitor based on the dry weight of the gauze (see 6.4). However, the contracting officer, at his discretion, may require that the gauze show no growth when tested as specified in 4.2.4.6.

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3.2 Physical properties. The filter material shall conform to table II when tested as specified in 4.2.4.

Table II. Physical requirements

Property	Requirement		
	Type 5	Types 6, 7, 8	Type 15-N
Thickness (inch)	0.019 to 0.030	0.033 to 0.043	0.045 to 0.055
Airflow resistance in mm. water (flowrate - 32 l/min)	19 to 27	44 max	27 max
DOP smoke penetration (percent) (flowrate - 32 l/min)	0.30 max	0.05 max	0.026 max
Tensile strength pound/inch width	----	2.0 min	1.0 min
Water repellency, inches	15 min	20 min	20 min
pH value	7.0 to 8.0	7.0 to 8.0	7.0 to 8.0
*Mildew inhibitor (percent)	2.5 of dry weight of furnish	2.5 of dry weight of furnish	2.5 of dry weight of furnish
Fold characteristics	----	No tears, breaks or cracks	----

*The contractor shall furnish a certificate giving the name, percentage, and manufacturer of the mildew inhibitor used. However, the contracting officer, at his discretion, may require that the filter material show no growth when tested as specified in 4.2.4.6.

3.3 Size. The filter material shall be wound into rolls as specified in 5.1. The width (including tolerance) of the rolls shall be as specified at the time of procurement (see 6.2).

3.4 Workmanship. The filter material shall contain no foreign matter, thick or thin spots, abrasions, tears, holes, punctures, or wrinkles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the filter material produced by one manufacturer, on one machine, from the same materials, under essentially the same manufacturing conditions, in no more than 24 consecutive hours.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.2.2.2 For tests. A test strip, six feet long for type 5, and three feet long for the other types, shall be taken from the free end of each slit roll across the entire width of the roll at a sufficient distance from the end to prevent inclusion of material damaged due to shipping and wrapping. Each test strip shall be identified with the roll from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination. Filter material shall be examined in accordance with the classification of defects (table III) and MIL-STD-105. Fifty consecutive yards shall be examined in each sample roll. The acceptable quality level shall be 15 major and 40 minor defects per 100 units (linear yards, not less than 36 inches wide).

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Table III. Classification of defects

*Defects	Classification	
	Major	Minor
Abrasion marks	X	
Cut, hole, or tear	X	
Too many splices		X
Splices not flagged		X
Wrinkle or crease		X
Foreign matter		X
Thick or thin spots	X	
Width incorrect	X	
Roll diameter incorrect		X

*Continuous defects shall be counted as one defect per linear yard.

4.2.3.2 For test. Each test strip of filter material shall be tested in accordance with 4.2.4. The acceptance number is zero.

4.2.4 Tests. Test specimens shall be taken at random from each test strip.

4.2.4.1 Thickness. The thickness of the filter material shall be determined using a dead weight type dial gage equipped with a dial graduated to read directly to 0.001 inch. The presser foot shall be circular with a diameter of 0.25 ± 0.01 inch. The presser foot and moving parts connected therewith shall be weighted so as to apply a total load of 3.0 ± 0.1 ounces to the specimen. The presser foot and anvil shall be plane and parallel to each other.

4.2.4.2 Airflow resistance and DOP smoke penetration. The airflow resistance and DOP smoke penetration of the filter material shall be determined on each test pad using the Q127 DOP Filter Testing Penetrometer (DLB136-42-102-5). A test pad shall consist of five layers for type 5 and one layer for types 6, 7, 8, and 15-N material. The test shall be performed at a flowrate of 32 liters per minute through an area of 100 square centimeters.

4.2.4.3 Tensile strength. The tensile strength shall be determined by method 171 of UU-P-31. In addition:

(a) The machine shall be calibrated to be read accurately, in the region of the test results, in 0.1 pound increments.

(b) Tests shall be on specimens cut in the paper-machine direction only.

(c) Test specimens shall be not less than 1 inch nor greater than 2 inches wide and shall be not less than 6 inches long.

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(d) Test specimens shall be conditioned as specified in method 102 of UU-P-31.

(e) Test results shall be reported in pounds per inch width of specimen.

4.2.4.4 Water repellency. The water repellency of the filter material shall be determined on each test pad with the Q101 Water Repellency Indicator (DLB125-8-1). The test pads shall consist of seven (7) layers for type 5 material and one layer for type 6, 7, 8, and 15-N material.

4.2.4.5 pH value. The pH of each roll of paper shall be determined immediately after treatment for water repellency as follows: Moisten a strip of short range, class B, indicator paper conforming to O-P-94, with distilled water. Place the moist indicator paper flat on the outer layer of the roll of paper. Within 10 seconds, check the color of the indicator paper in accordance with O-P-94.

4.2.4.6 Mildew inhibitor. The mildew resistance of the filter material and gauze backing shall be determined using method 5751.1 of CCC-T-191. Evaluation shall be qualitative.

4.2.4.7 Fold characteristics. A fold characteristic test shall be performed on specimens of types 6, 7, 8, and 15-N filter material. Each specimen cut 6 inches long in the machine direction and 3 inches wide, shall be bent across the machine direction through an arc of 180° over a 3/16-inch-diameter round mandrel.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. The filter material shall be wound evenly on a hollow core, constructed of paper or other suitable material, having an inside diameter of 3-1/16 + 1/8 inches. A maximum of four splices is permitted for each roll. The location of the splices shall be indicated by a paper "flag" having a contrasting color. Each roll shall have a maximum outside diameter of 25 inches and not more than 3 percent of the rolls in a lot shall have an outside diameter of less than 20 inches. No splices shall be permitted on the roll in that portion exceeding 20 inches in diameter. The rolls shall be wrapped with two thicknesses of water-proof barrier material conforming to type C-1 of PPP-B-1055. The wrapper shall be closed at the ends by means of inside and outside headers, using sufficient adhesive to make water tight seals. The barrier material shall be sealed with an adhesive conforming to type II, grade B of MMM-A-260.

5.1.1 Packaging, level B. The filter material shall be packaged the same as that specified in paragraph 5.1.

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5.1.2 Packaging, level C. The filter material shall be packaged to provide adequate protection against corrosion, moisture, dust, fungus, deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level A. The filter material, packaged as specified in 5.1, shall be packed in wood boxes conforming to style A of PPP-B-621. The net weight of the contents of each box shall not exceed 250 pounds. The boxes and the finished wood parts thereof, shall be completely immersed for a minimum of three minutes in a water repellent wood preservative conforming to type II, composition A of TT-W-572. Prior to packing the item in treated wood box, the preservative dryness must meet the requirements of MIL-B-2427.

5.2.1 Packing, level B. The filter material shall be packaged the same as that specified in 5.2, except no wood preservation will be applied.

5.2.2 Packing, level C. The filter material, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Shipping containers shall be in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with D150-2-1. All markings on the shipping containers shall be clear, legible, and permanent after application of the wood preservative specified in 5.2.

5.3.1 Roll identification. Each slit roll shall be marked with the lot number, and the reel number, and the location in the reel from which it was taken in the following manner.

(a) Each reel in the lot shall be marked consecutively starting with number 1. All rolls rewound from one reel shall have the same number as the reel.

(b) Each roll rewound from one reel shall be marked alphabetically, with the letter A indicating the first roll rewound from the reel. Successive rolls from the same reel shall be marked B, C, etc.

(c) In addition, slit rolls shall be numbered consecutively, with number 1 indicating the "front position" on the machine. (The "front position" is that side of the machine from which the machine tender controls the sheet as it progresses toward the reel.)

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6. NOTES

6.1 Intended use. This specification covers five types of filter material intended for use as the filter medium for removing particulates from influent air in gas masks and collective protectors.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Type of filter material required.
(d) Width (including tolerance) of the rolls to be supplied.
(e) Whether or not the contractor is to certify the amount and type (company and trade designation) of mildew inhibitor used.

6.3 Gauze backing. The thickness requirement for type 5 material has been met by using gauze backing having 12-20 threads per inch in the wrap and 11-18 in the fill. However, any other gauze may be used which will enable the finished paper to meet the thickness requirement of this specification.

6.4 Mildew inhibitor. The following mildew inhibitors have been found satisfactory:

Shirlan A
Zerlate (Milban D)

Custodians:

Army - MU
Navy - BuShips

Preparing activity:

Army - MU(EA)

Project No. 4240-A412

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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